



$$\frac{1}{3} + \frac{1}{15} + \frac{1}{35} + \cdots + \frac{1}{35 \times 37} = \underline{\hspace{2cm}}$$
$$\frac{1}{1 + \sqrt{2}} + \frac{1}{\sqrt{3} + \sqrt{2}} + \frac{1}{\sqrt{4} + \sqrt{3}} + \cdots + \frac{1}{\sqrt{2019} + \sqrt{2020}} +$$
$$\frac{1}{\sqrt{2020} + \sqrt{2021}} + \frac{1}{\sqrt{2021} + \sqrt{2022}} = \underline{\hspace{2cm}}$$
$$\sqrt{6 + 3\sqrt{3}} + \sqrt{6 - 3\sqrt{3}} = \underline{\hspace{2cm}}$$
$$A = \frac{1^2 + 2^2}{1 \times 2} + \frac{2^2 + 3^2}{2 \times 3} + \frac{3^2 + 4^2}{3 \times 4} + \cdots + \frac{1003^2 + 1004^2}{1003 \times 1004} +$$
$$\frac{1004^2 + 1005^2}{1004 \times 1005}$$
$$A = \underline{\hspace{2cm}}$$
$$\frac{1}{1 \times 2 \times 3} + \frac{1}{2 \times 3 \times 4} + \cdots + \frac{1}{98 \times 99 \times 100} = \underline{\hspace{2cm}}$$
$$x^3 + 2x^2 + 2x + 1 = 0 \quad x^{1994} + x^{1997} + x^{2000} = \underline{\hspace{2cm}}$$
$$a, b, c \quad x = a^2 - 2b + \frac{\pi}{3} \quad y = b^2 - 2c + \frac{\pi}{6} \quad z = c^2 - 2a + \frac{\pi}{2}$$
$$x, y, z \quad \frac{1}{x} + \frac{1}{y} = 2 \quad \frac{2x + 3xy + 2y}{x + 2xy + y} = \underline{\hspace{2cm}}$$

$$\begin{cases} \frac{1}{3-\frac{x}{2}} + \frac{1}{y-\frac{2}{5}} = 2 \\ \frac{1}{3-x} - \frac{1}{y-2} = -3 \end{cases} \quad 17x^2 + 10y^2 = \underline{\hspace{2cm}}$$

$$\frac{x}{x^2-x-1} = 4 \qquad \frac{x^2}{x^4+x^2+1} = \underline{\hspace{2cm}}$$

$$a,b,c \qquad \frac{ab}{a+b} = \frac{1}{3} \quad \frac{bc}{b+c} = \frac{1}{4} \quad \frac{ca}{c+a} = \frac{1}{5} \qquad \frac{abc}{ab+bc+ca}$$

$$\frac{1}{6} \qquad \frac{1}{12} \qquad \frac{1}{15} \qquad \frac{1}{20}$$

$$\frac{x^2+6y^2-10z^2}{3x^2-4yz+5z^2} = \frac{m}{n} \qquad xyz \neq 0 \qquad x+2y+z=0 \quad 5x+4y-4z=0$$

$$m,n \qquad 10m+n = \underline{\hspace{2cm}}$$

$$\frac{1}{x} + \frac{1}{y} + \frac{1}{z} + 600 \qquad \underline{\hspace{2cm}} \qquad x,y,z \qquad \sqrt{x+y} + \sqrt{y+z} = \sqrt{z+x}$$

$$a^2+b^2+c^2 = (a+b+c)^2 \qquad \frac{b+c}{a} + \frac{c+a}{b} +$$

$$\frac{a+b}{c} + 100 = \underline{\hspace{2cm}}$$

$$abc=1 \quad a+b+c=2 \quad a^2+b^2+c^2=3 \quad \frac{1}{ab+c-1} +$$

$$\frac{1}{bc+a-1} + \frac{1}{ca+b-1} \qquad -1 \qquad -\frac{1}{2} \qquad 2 \qquad -\frac{2}{3}$$

$$x = \frac{1}{\sqrt{2}-1} \quad a \quad x \qquad b \quad -x \qquad a^3+b^3+3ab$$

$$4\sqrt{2}+2 \qquad 4\sqrt{2}+3 \qquad 2 \qquad 1$$

$$a^2+1=3a \quad b^2+1=3b \qquad a \neq b \qquad \frac{1}{a^2} + \frac{1}{b^2}$$

$$\frac{a}{b} = a+2 \quad \frac{a}{b} = a-2 \qquad \frac{b^2}{(a-2)^2} = \underline{\hspace{2cm}}$$

$$\frac{a}{b} \qquad ab = a+2 \quad \frac{a}{b} = a-2 \qquad \bullet$$

$$m=\frac{2020}{\sqrt{2021}-1}\qquad m^5-2m^4-2020m^3+m^2-2m-2021\qquad \rule{1cm}{0.4pt}$$

$$\begin{array}{ccccc} & & a,b,m & a+b=m+2 & ab=4m \\ m & & & & \\ & \rule{1cm}{0.4pt} & & & \\ & n & n^2+n+51 & & n \end{array}$$

$$\begin{array}{ccccc} & & & & \\ & & & & \\ \rule{1cm}{0.4pt} & & & & \\ a & a=\sqrt{1-\frac{1}{a}}+\sqrt{a-\frac{1}{a}} & [a]= & & [a] \\ & a & & & \end{array}$$

$$\begin{array}{ccccccc} & & 1,3,6,10,\ldots & & & & \\ 1,4,9,16,\ldots & & & & & & \\ & 4=1+3 & 9=3+6 & 16=6+10 & & & \\ 21+28 & 20=6+14 & 25=9+16 & 36=16+20 & 49= & & \end{array}$$

$$\begin{array}{ccccccc} & & 1\times 2\times 3\times\cdots\times n & n! & 5!=1\times 2\times 3\times 4\times 5 & & \\ 100!=1\times 2\times 3\times\cdots\times 100 & & S=1\times 1!+2\times 2!+3\times 3!+\cdots+2007\times & & & & \\ 2007! & S & & & & & \end{array}$$

$$\begin{array}{ccccccc} & & m & n & m,n & & \\ & m=674,n=1346 & m=674,n=1347 & m=675,n= & & & \\ 1348 & m=675,n=1349 & & & & & \end{array}$$

$$\begin{array}{ccccccc} & & A & C & & & \\ & & & & & & \\ & AB & BC & CD & DA & & \\ \hline \end{array}$$

$$\frac{7}{13} < \frac{n}{n+k} < \frac{6}{11}$$

n, k

n

k

n
